

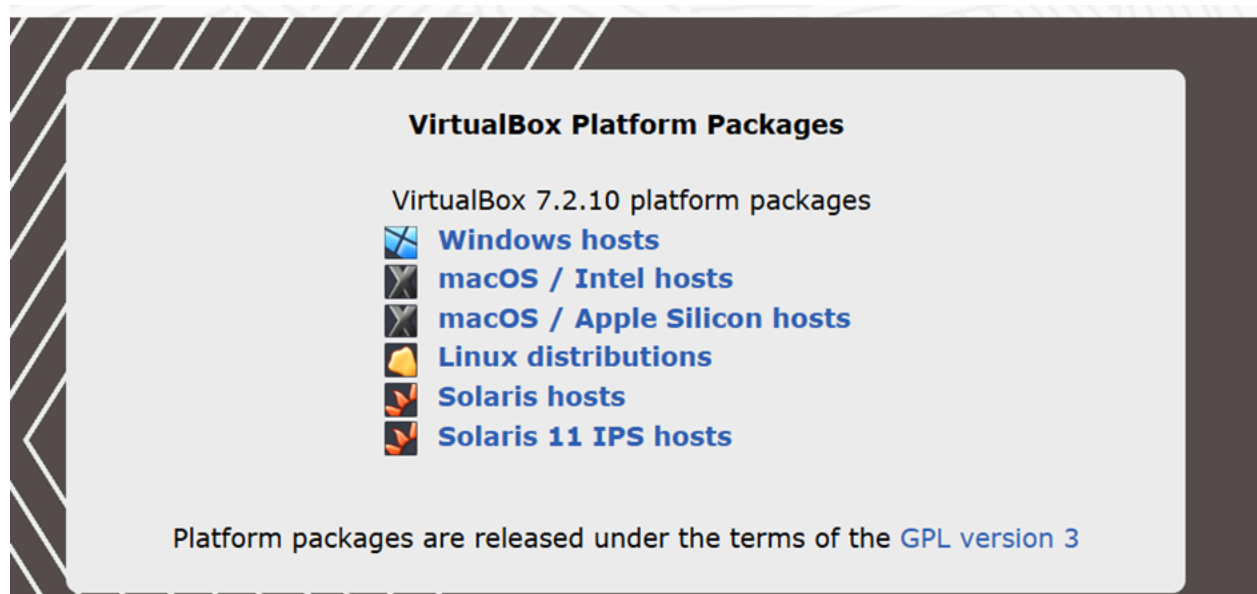
Introduction

The inspiration for this project originated from the procedures I learned during my time at my PC Services internship. The internship taught me how to manage endpoint users using Active Directory (AD) and Microsoft Configuration Manager (used to be called SCCM). I decided I wanted to create a virtual lab using VirtualBox in order to practice my skills in AD. During my journey completing this project, I learned a lot about endpoint management, virtual machines, and some networking concepts like DHCP and more. For the domain controller, I loosely followed a tutorial on how to set it up, but before I followed each step, I wanted to try it by myself first so that I could test my knowledge and struggle on my own. The set-up process definitely humbled me, and I had to do some extra research to help me troubleshoot any issues I came across. Everything else was done by trial, error, prior knowledge gained from my internship, and a lot of googling.

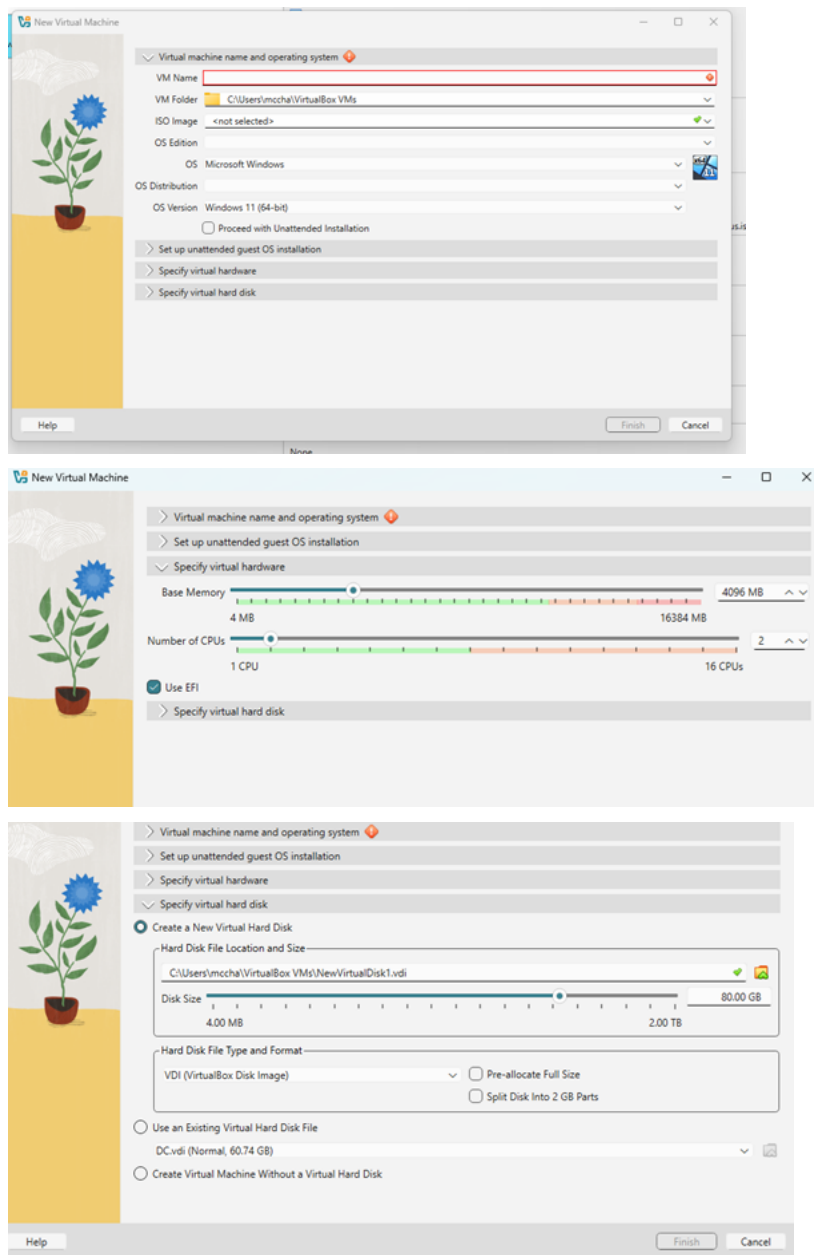
I documented my process and wanted to share it.

The Setup

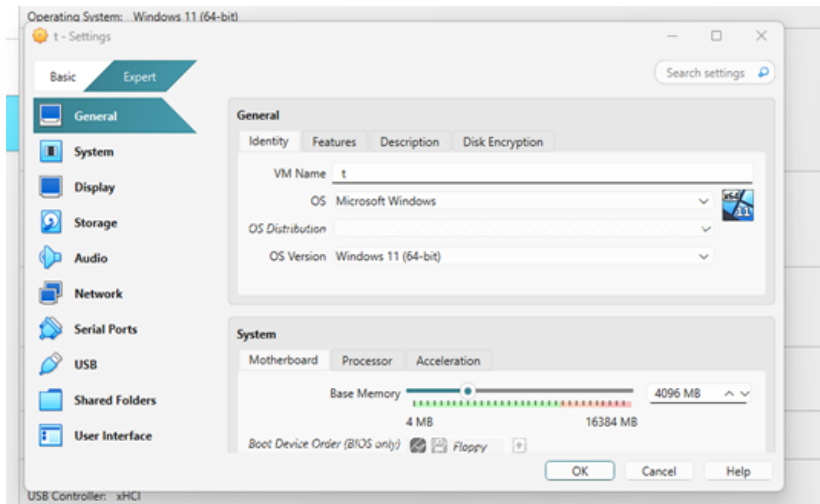
First, I downloaded Oracle VirtualBox (<https://www.virtualbox.org/>) and selected the “Windows hosts” platform package.



I found that the process to create a basic virtual machine was relatively straightforward. It is similar to a physical computer in the context that you need to allocate a certain amount of storage, RAM, CPU cores, disk size, etc..



When customizing a VM, the settings tab (especially the “Expert” tab) looked a little daunting but after giving it a good look, and researching things that I didn’t know, I realized that I didn’t need to touch a lot of the options. Thankfully, most of the default options were things that I needed in order to just create a basic VM.



Next, I wanted to download the correct operating system for my virtual machine. Since I didn't want an older version of Windows, I downloaded the Windows 11 disk image from Microsoft's Software Download page. Specifically, I used the "Download Windows 11 Disk Image (ISO) for x64 devices" section. To make things easy, I just downloaded the current release which was Windows 11 version 25H2. This can be found at:

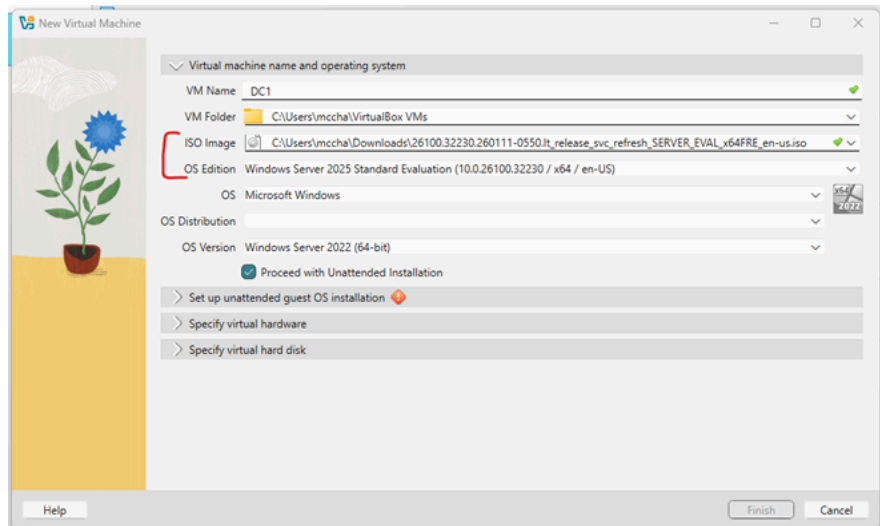
<https://www.microsoft.com/en-us/software-download/windows11>

After doing a tiny bit of research (asking google what else I needed), I found that you must also download Windows Server in order to access AD. I downloaded Windows Server 2025. It can be found at: <https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2025>

Setting up the Domain Controller VM and AD

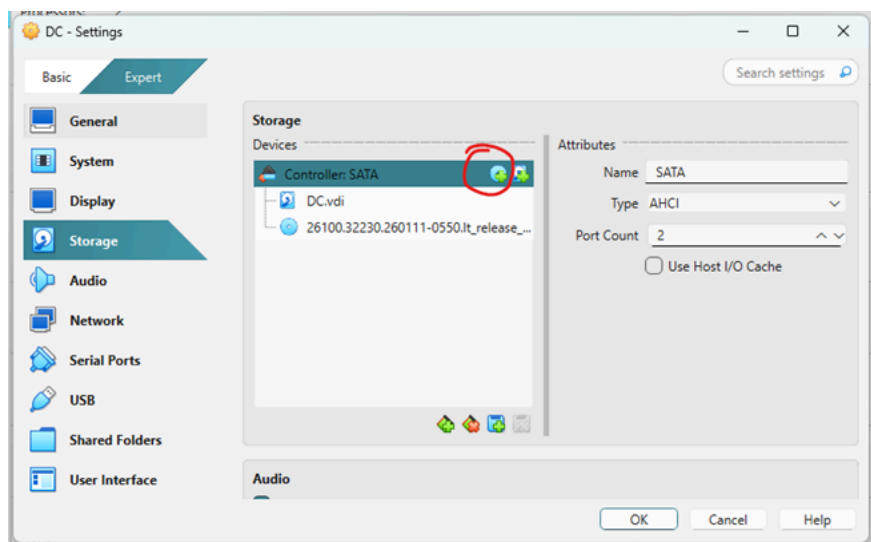
Now I had all that I needed to create my Domain Controller VM which would have the Windows Server 2025 disk image installed, and the client VMs which would have the standard Windows 11 disk image.

After opening VirtualBox, I clicked on "New" then filled in the fields that I knew, as well as added sufficient virtual hardware based on my host machine's capabilities.

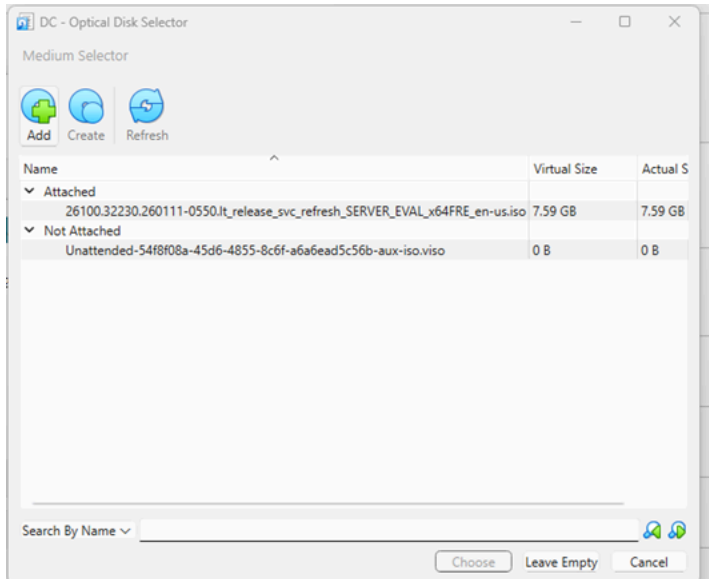


(Added 4GB of memory, 2 CPU cores, 60 GB of Storage)

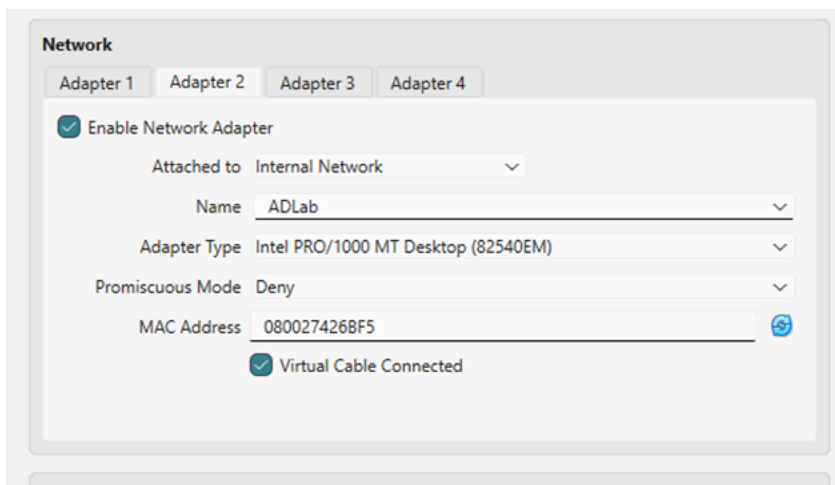
Next, I clicked finish then settings. I switched to the “Expert” tab, then “Storage”. Clicked on the “Add optical drive” option and input the Windows Server ISO.



Then clicked “Add” and browsed my folder to find the ISO image.



After that, I went to the “Network” tab and kept the first adapter the same, then went to the “Adapter 2” tab, enabled it, and set it from “NAT” to “Internal Network”. I named it ADLab.



Finally, I booted up the VM.

I then followed the instructions on the screen and eventually clicked to install the “Windows Server 2025 Standard Evaluation (Desktop Experience)”. After finishing the installation, I was asked to create an admin password. Finally, I was brought into the Server Manager.

I did some more research and was told to set up the IP address and DNS. Went into Settings → Network & Internet → Ethernet → IP assignment. I put the private IPv4 address starting with

192.168.10.10. I learned that it is a factory default address to access their admin dashboard.
(<https://192-168-1-1ip.mobi/192-168-10-10/>)

IP Address: 192.168.10.10

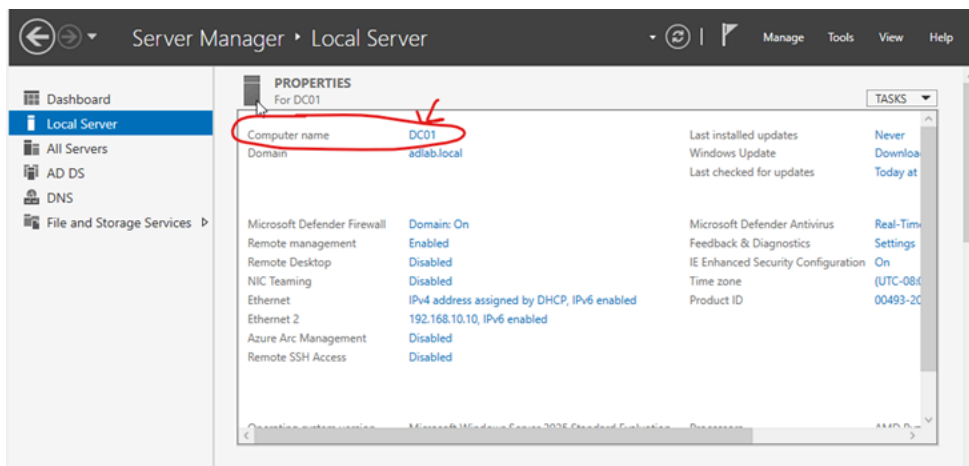
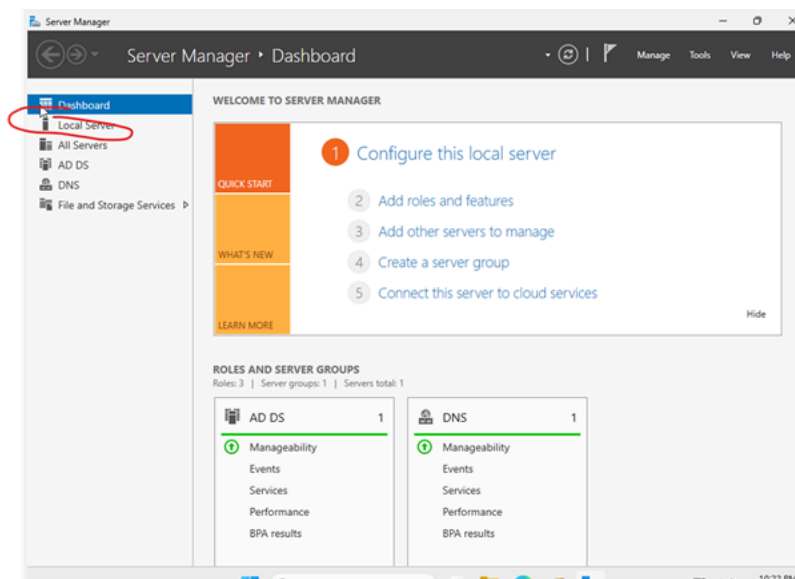
Subnet Mask: 255.255.255.0

Default Gateway: (left blank)

Preferred DNS: 192.168.10.10

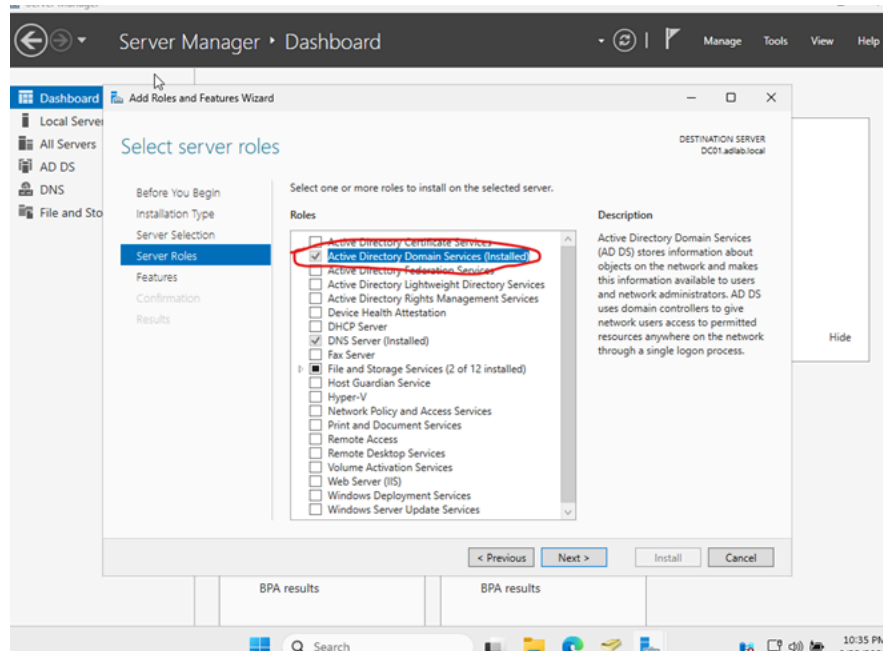
(***Note: input this information into the second adapter instead of the NAT adapter)

Next, I went back into Server Manager and renamed the server. Clicked on the old computer name and renamed it to DC01.

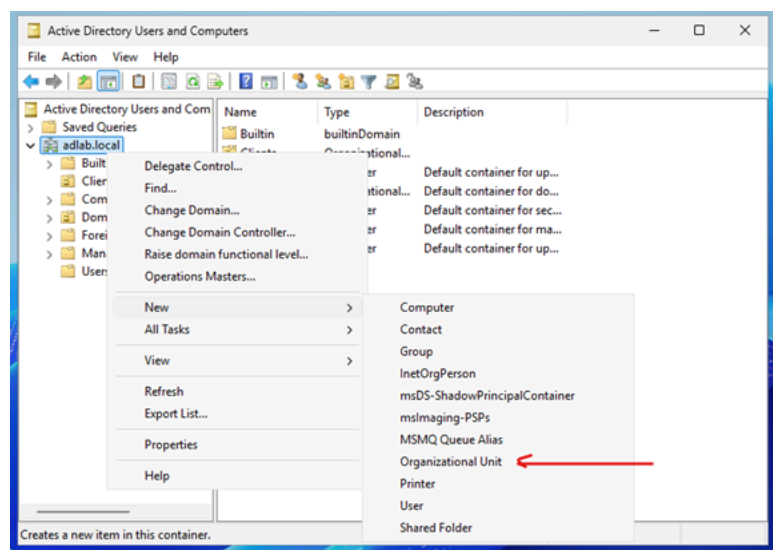


After renaming and restarting the server, it was time to install AD. In the server manager, I went to Manage → Add roles and Features → and clicked next until I reached Server Roles (will take

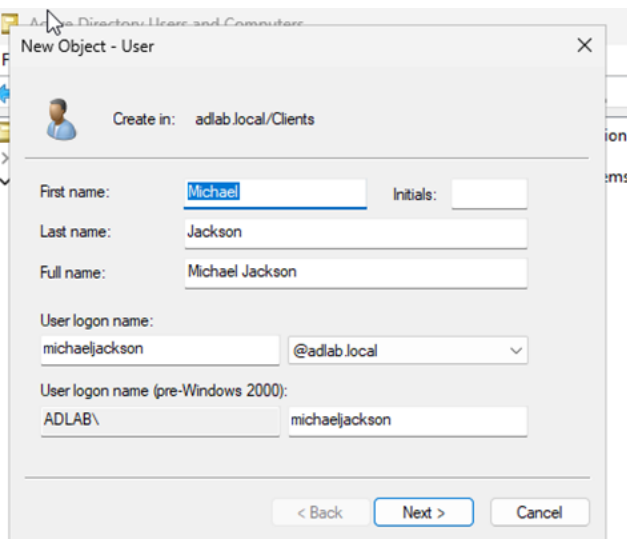
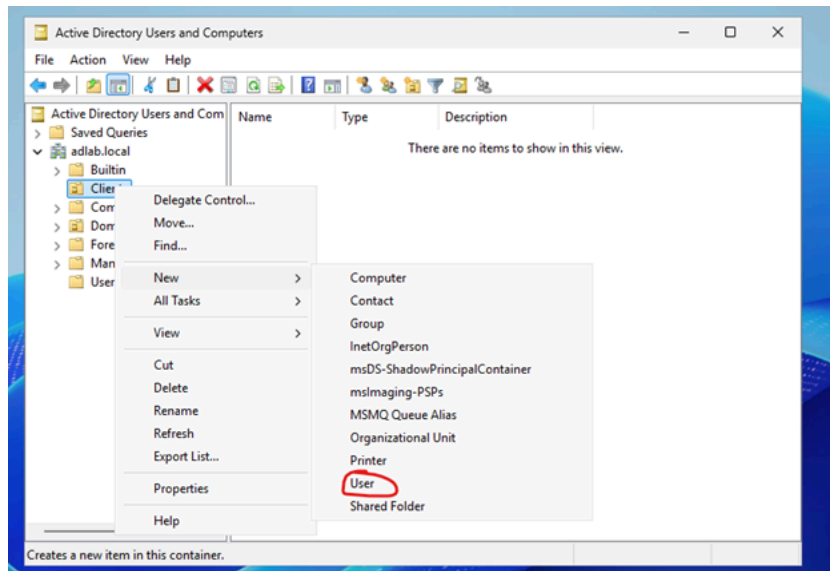
a moment). I clicked Active Directory Domain Services and followed the remaining steps. When that installation finished, I promoted the server to a Domain Controller by clicking on the flag notification and hit “promote this server to a domain controller” → “add a new forest” → then typed in my domain name “adlab.local”. Then, clicked Next, left DNS Server and Global Catalog checked, entered a DSRM password (and wrote it down), then “Next” through the remaining screens until I was able to click Install.

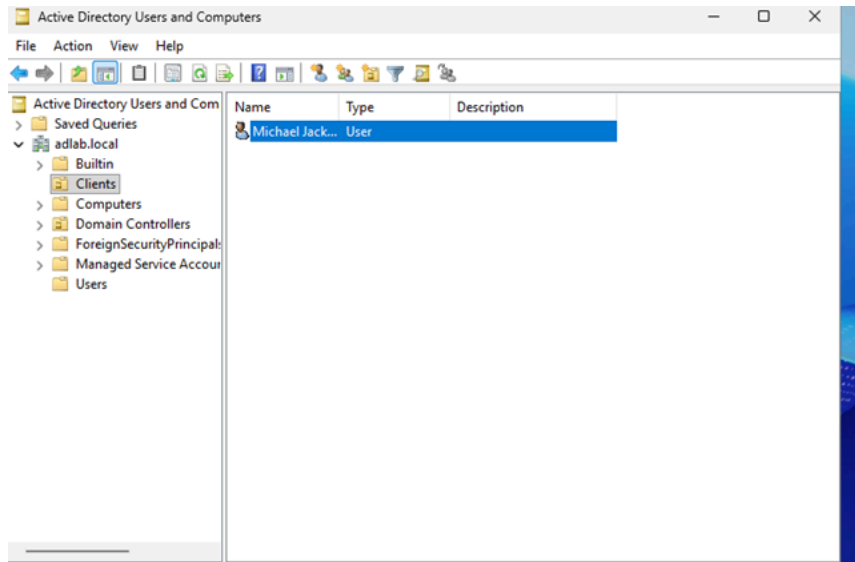


The server eventually restarted and then I logged in as administrator to check if AD was properly installed and was working. I then created an Organizational Unit (OU) called “Clients” by right clicking the domain name → New → Organizational Unit.



I then decided to start messing around with AD a little bit. I created a new user called Michael Jackson.

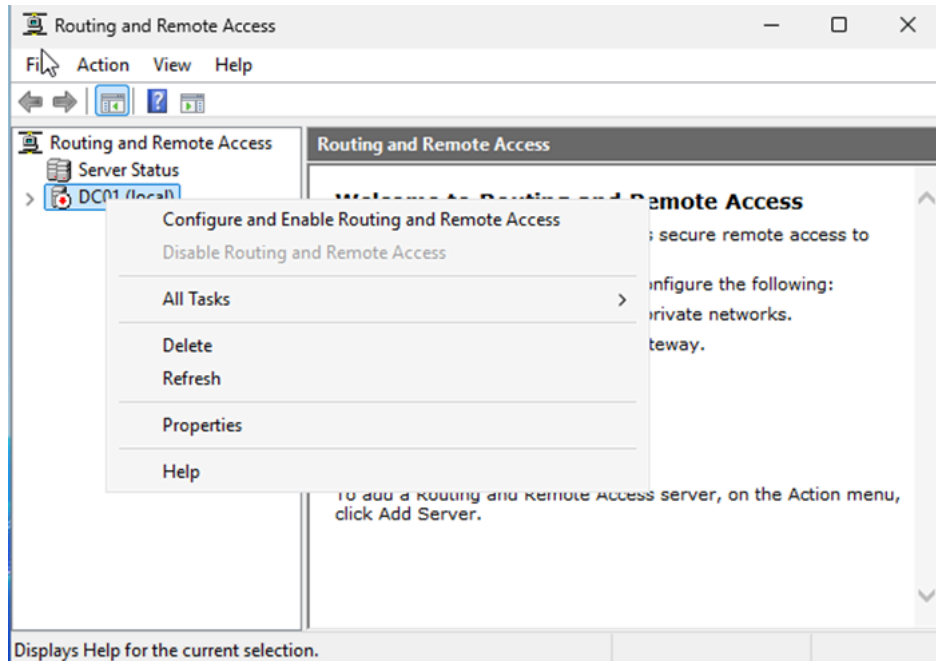




Routing and Remote Access

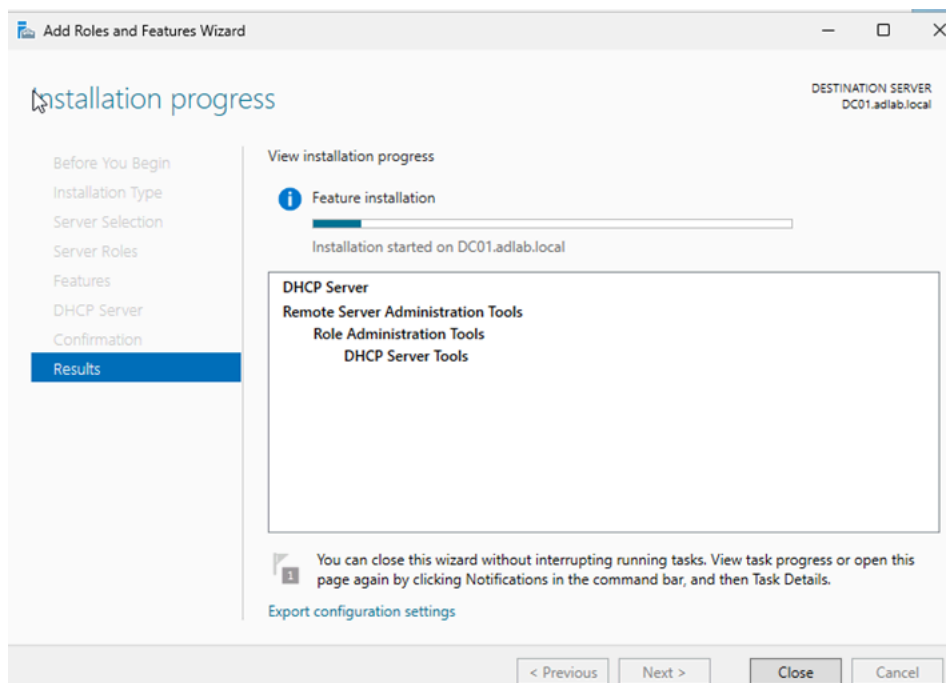
Next, I set up Remote Access so the client VM can connect to the internet through the DC similar to a VPN. On the Server Manager, I clicked “Add roles and features” and clicked next until I got to the “Server Roles” tab. I selected Remote Access and clicked next. I then checked the routing box and then “Add Features” at the bottom of the pop-up. Then I clicked next and finished the installation.

After that finished, I went to Tools → Routing and Remote Access → right clicked on the domain name → “Configure and Enable Routing and Remote Access. Clicked on the NAT (Network Address Translation) option and chose the Ethernet that was internet facing (Adapter 1).

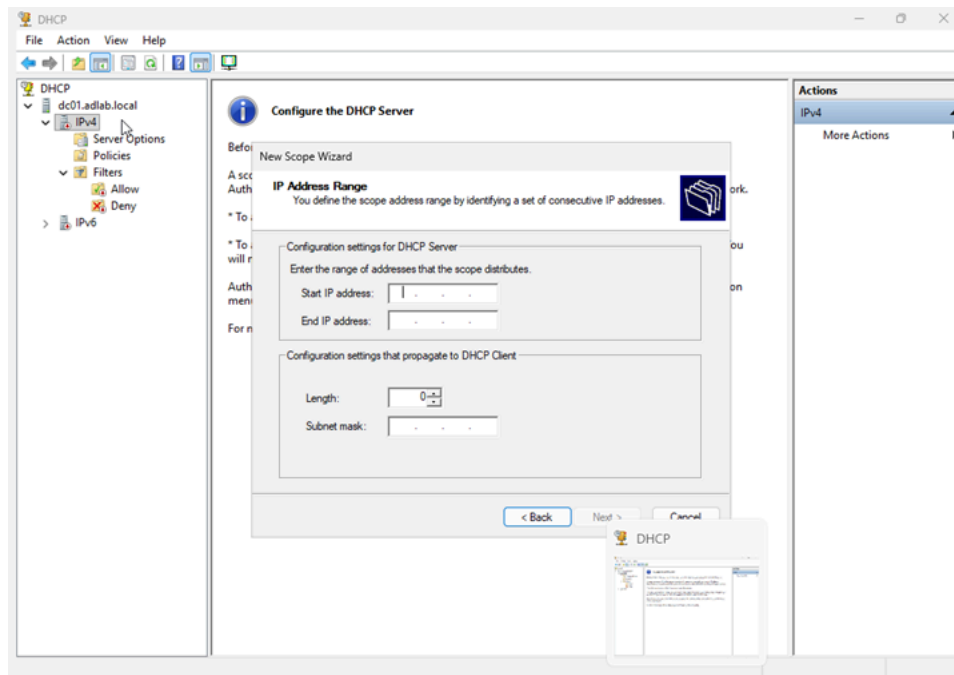


DHCP Server

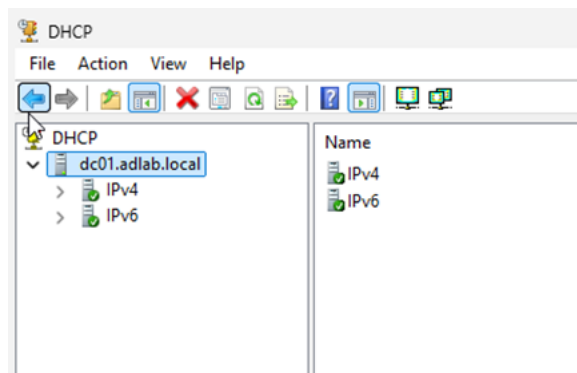
I then returned to “Add roles and features” in the Server Manager and selected DHCP Server which is in the Server Roles tab. The Installation process is the same as the Remote Access step.



When that finished, it was time to configure it. I went back to the Tools tab in Server Manager and clicked on DHCP. Clicked on my domain name and right-clicked “IPv4” and selected New Scope. I set the Start and End IP addresses to 192.168.10.100 – 192.168.10.200. This means that any client that joins the domain will be assigned an IP within that range. I set the subnet mask to 255.255.255.0. (**Note: I updated the scope to fix my troubleshooting process. Check the “Troubleshooting DHCP” tab near the bottom of the document for more details.)

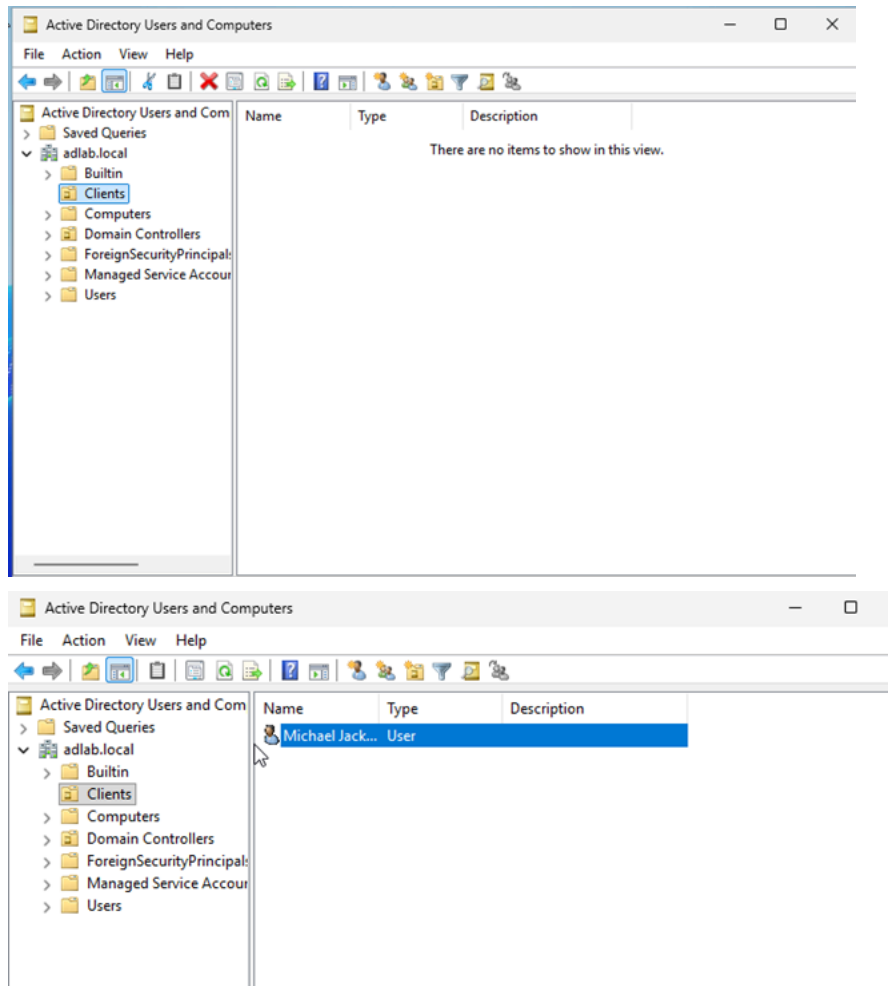


I clicked next until I hit the Routing page. I added 192.168.10.10 as the default gateway and clicked next. Finally, I right-clicked the domain name → clicked “Authorize” → right-clicked the domain name again → “Refresh”.



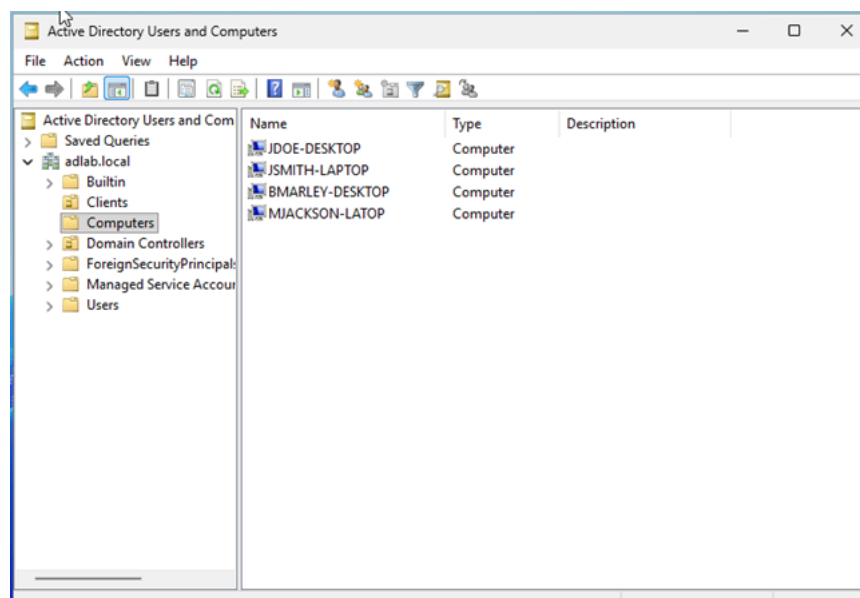
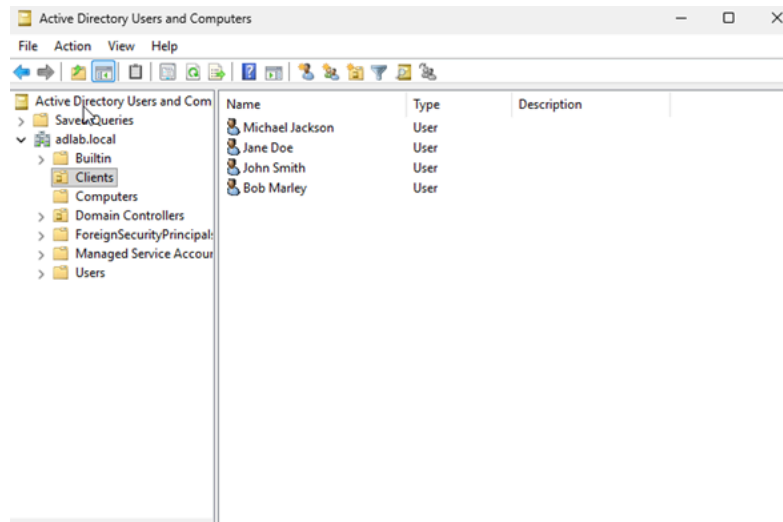
Adding Users to AD

I found the process of adding users and computers with AD easy and straightforward. That may be due to my little knowledge of using AD from the internship I'm working at right now. Once I opened AD, I right clicked the domain name and added an organizational unit (OU) called "Clients". I then added a user in that OU called "Michael Jackson" and created his username and login for his account.



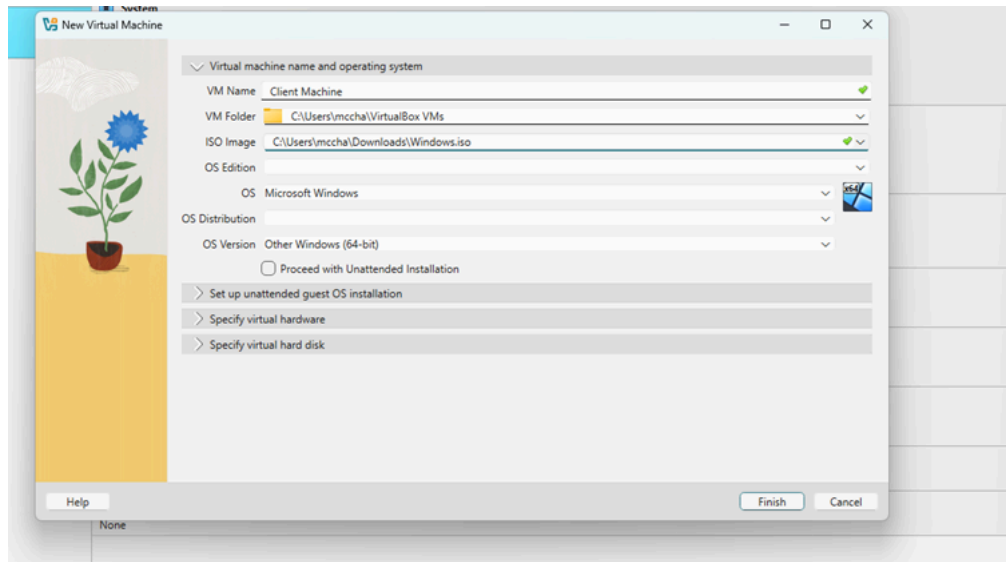
Just for fun, I added some more users later on. However, I'm only going to be setting up a VM for the Michael Jackson user right now.

Next, I navigated to the "Computers" container and added some fake computers just to fill up some space. (I will set up Michael Jackson's computer in the next section).



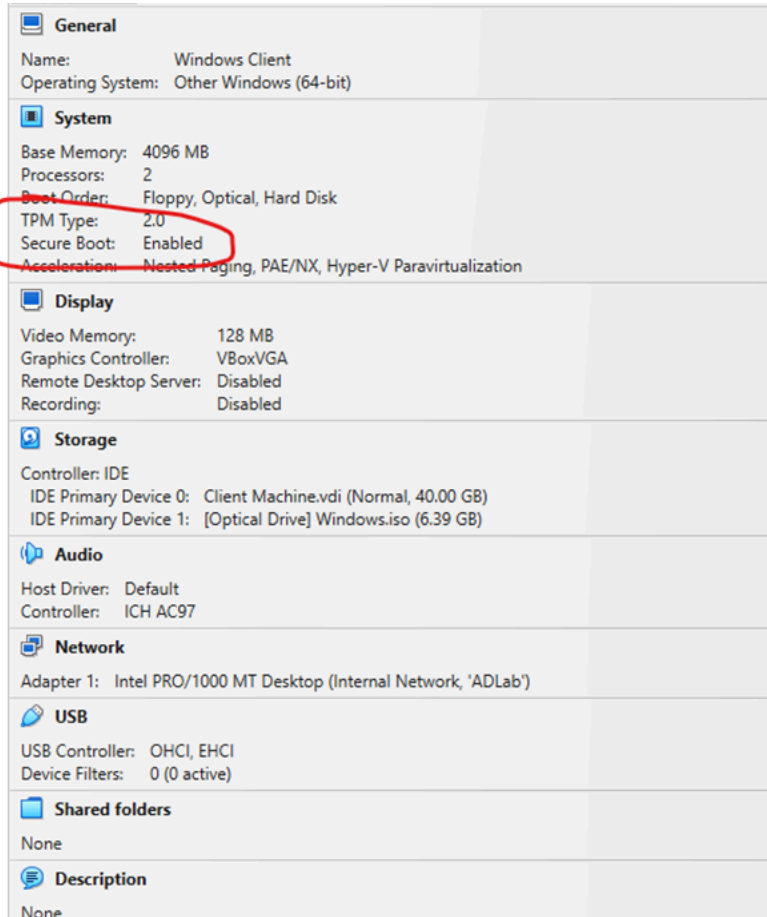
Setting up the Client VM

Setting up the VM was a lot simpler than the Domain Controller. All I needed was the Windows 11 ISO file and add the virtual hardware to the machine.



(*Notice in the Network section, I selected “Internal Network” and set it to the same name as the second adapter in my DC machine. Also, in the “Features” tab of the general section, I’ve changed the shared clipboard and drag-and-drop options to “Bidirectional”, just like we did in the DC VM.)

Next, I started the VM and followed the instructions to install Windows 11 on the client VM. I clicked on Windows 11 Pro to install. The first time I tried installing, I was prompted to make sure my PC supported TPM 2.0, 4GB of memory, and Secure Boot. I then shut down the VM and changed those options in the settings.



After a long installation process, I was finally able to boot up my Windows 11 VM for a user. I was prompted to update the computer to have the latest security updates, so it took even longer, unfortunately.

Joining the Domain

This process had me scratching my head a little bit because the original method I tried did not work. I originally just configured the IP address (similar to that of the DC VM) and tried to connect to the domain that way. What I didn't realize was that I set up my IP address incorrectly and the DC was not able to talk to my Client VM. In order to fix this issue, I made sure that the

network adapters on both VMs were connected (was also an issue because originally, they were not all connected), and the IP was configured correctly (I set 192.168.10.10 to be the IP for the internal network of the DC). I then pinged the DC IP address on the Client VM to make sure that they could see each other on the network.

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

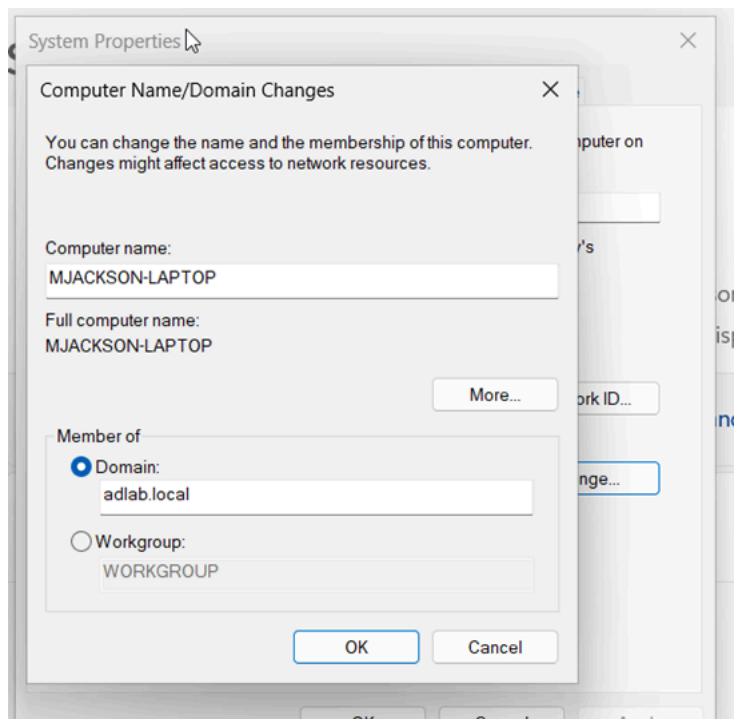
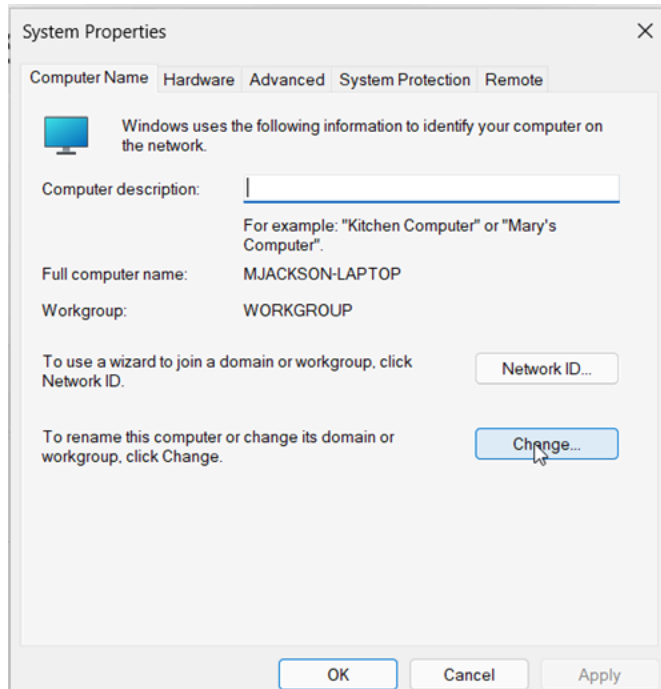
C:\Users\Michael Jackson>ping adlab.local

Pinging adlab.local [192.168.10.10] with 32 bytes of data:
Reply from 192.168.10.10: bytes=32 time=2ms TTL=128
Reply from 192.168.10.10: bytes=32 time=2ms TTL=128
Reply from 192.168.10.10: bytes=32 time=2ms TTL=128
Reply from 192.168.10.10: bytes=32 time=4ms TTL=128

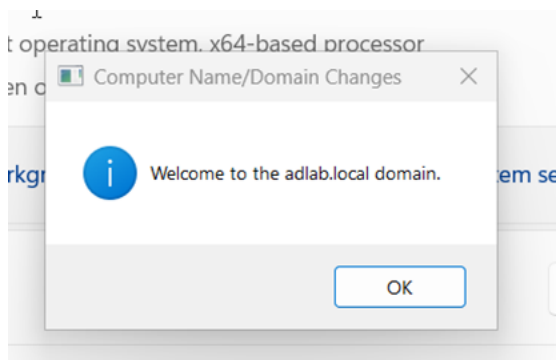
Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Users\Michael Jackson>
```

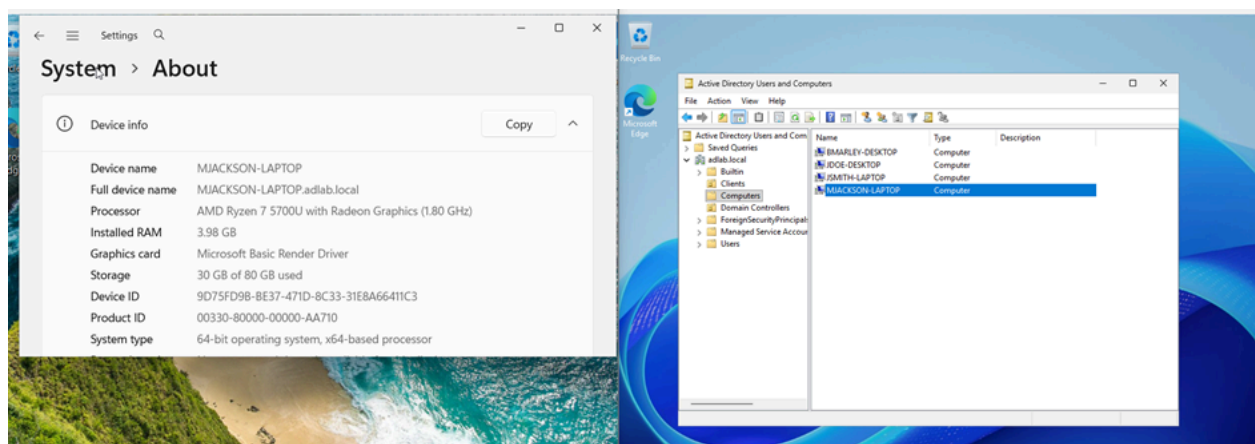
After I got a response from the DC VM, I connected the Client VM to the domain.



And after putting in the administrator login (adlab.local\administrator and password)

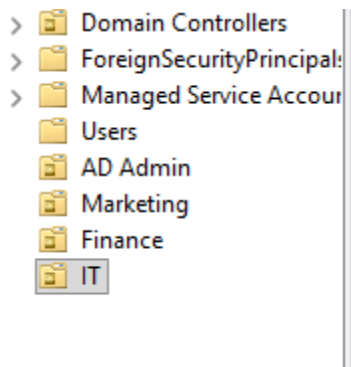


Our Client (Michael Jackson) is now in our domain!



Creating Organizational Units (OU) and Moving Users Around

I wanted to play around with creating more users and moving them into certain groups. I wanted to sort of simulate a corporate environment where the OUs were named example departments of a company.



I then created a few more users and moved them and their computers into the different OU's.

AD Admin – Michael Jackson

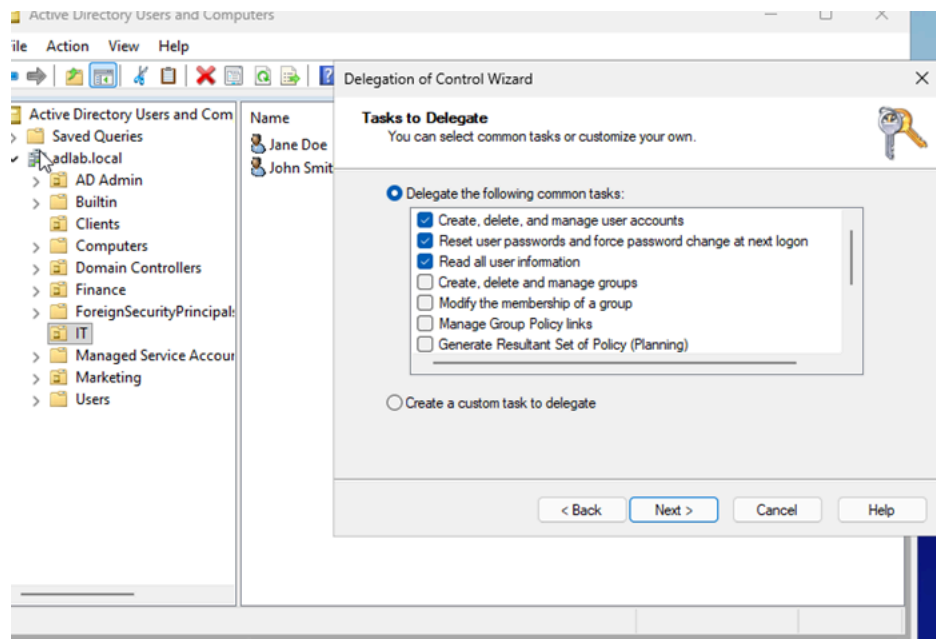
Marketing – George Washington & Lebron James

Finance – Lionel Messi & Bob Marley

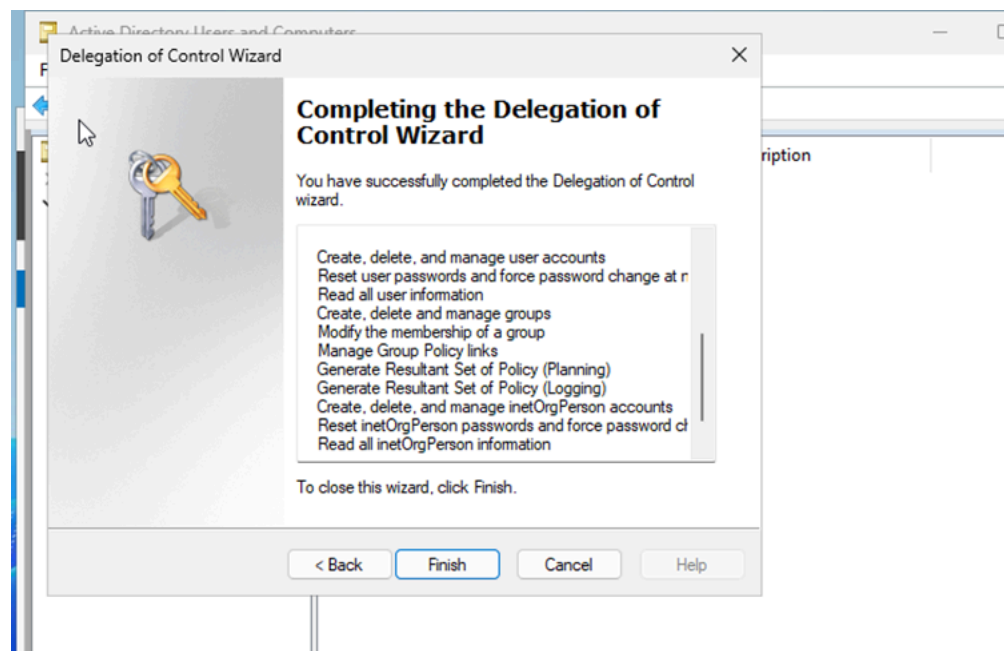
IT – John Smith & Jane Doe

Delegation Of Control

I selected the IT OU and clicked on the delegate control option. I then added both the users that are in the IT OU (Jane Doe and John Smith) and allowed them to do the following features:

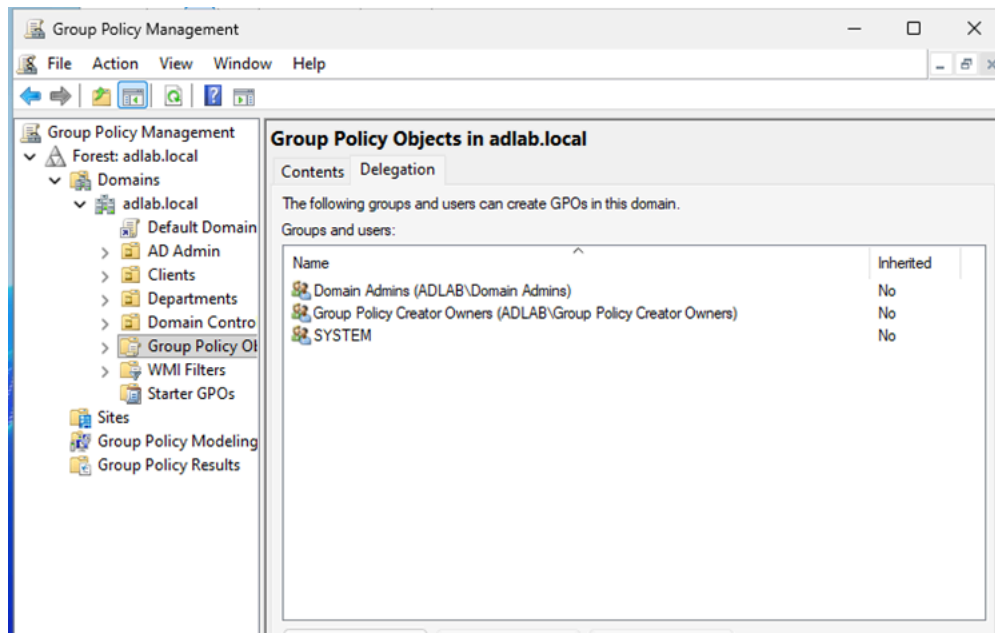


I then did the same thing for the AD Admin (Michael Jackson) except I allowed them to have control over all of the options.

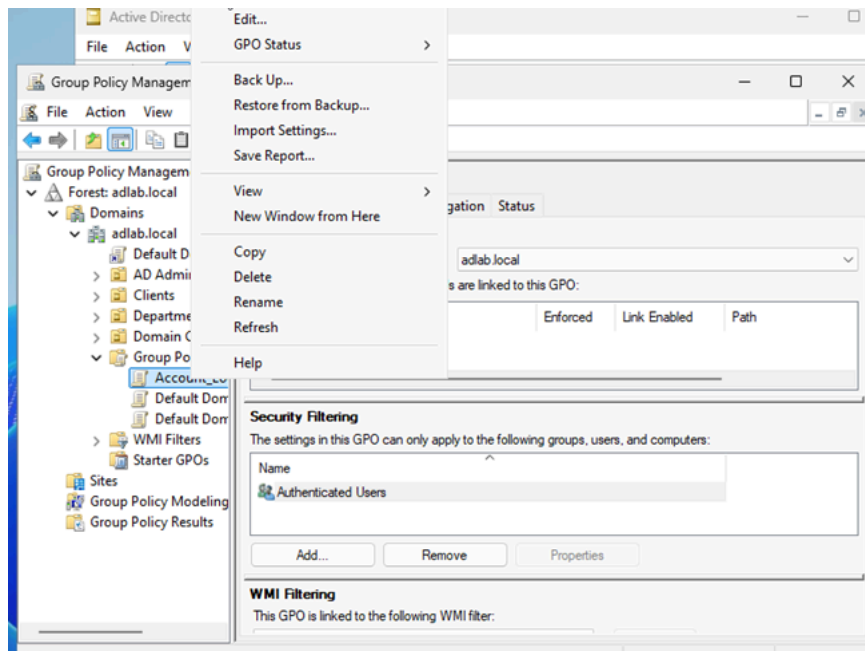


Group Policy Management

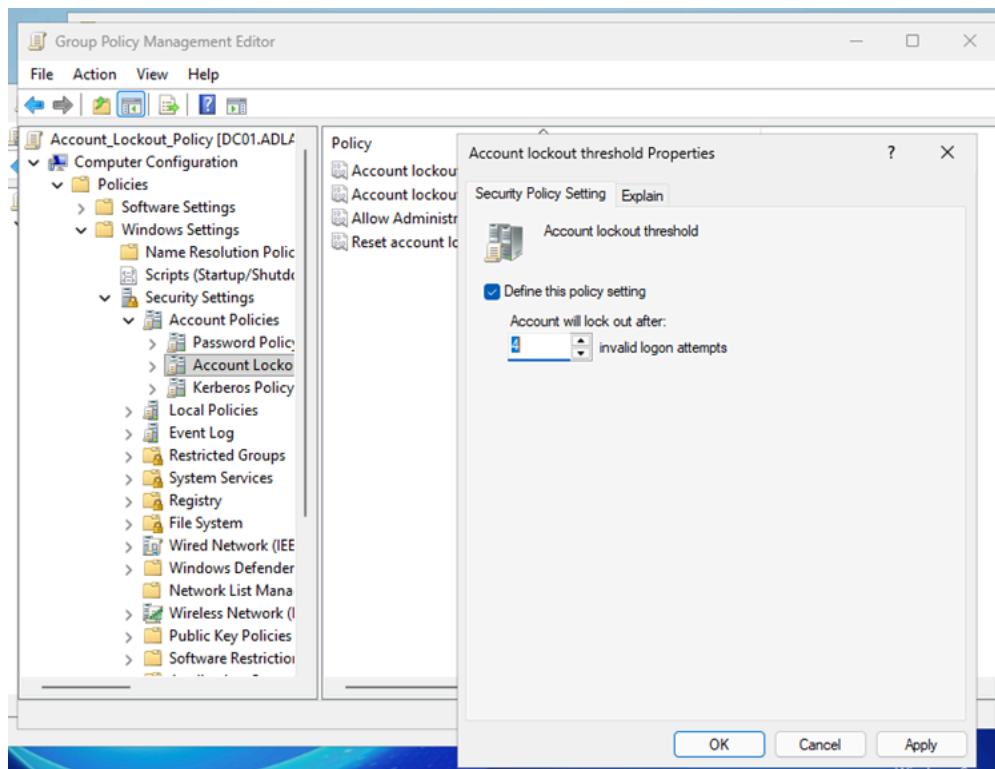
Group Policy Management is one of the big things I wanted to practice in this lab because I've heard so much about it when talking about the things you can do with Active Directory. After opening the Group Policy Management app on my DC VM, I then found the Group Policy Objects folder.



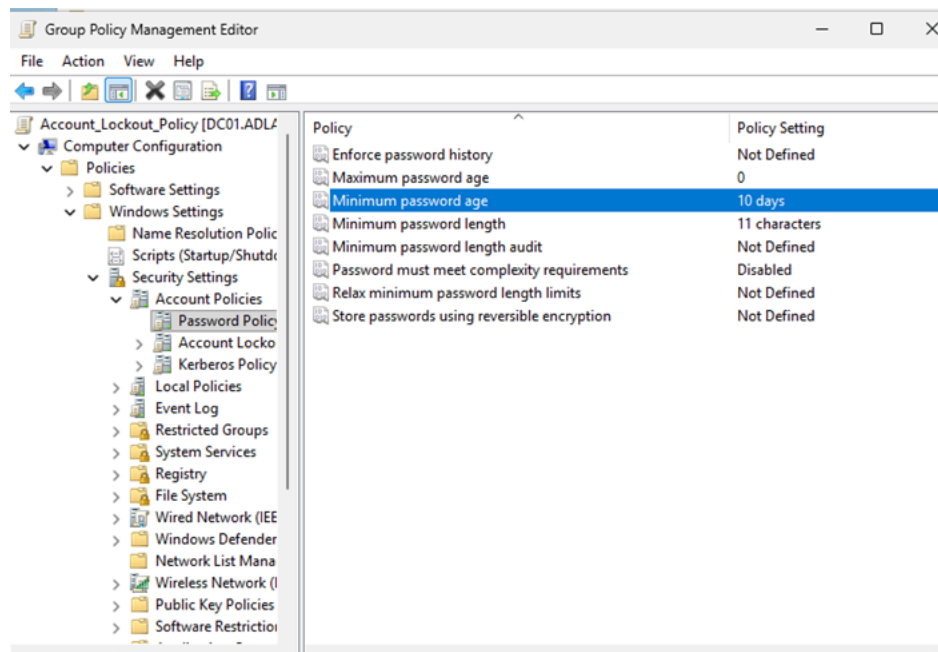
After right-clicking the Group Policy Objects folder and clicking new, I created an "Account_Lockout_Policy" and then expanded the GPO folder and right clicked the new GPO.



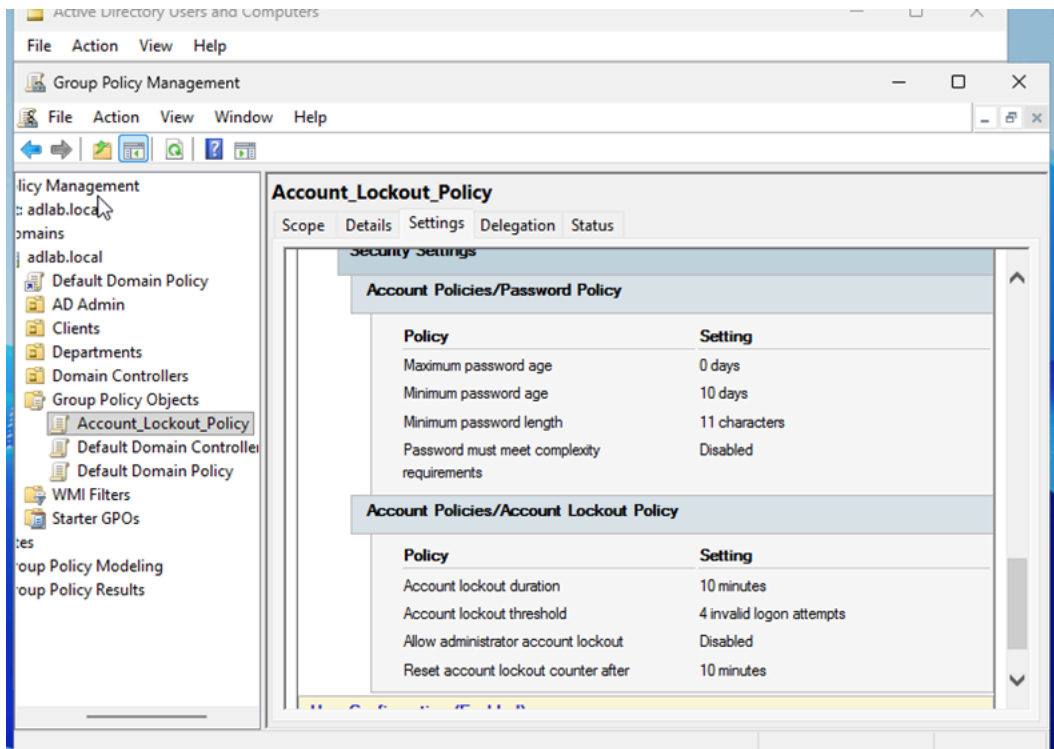
I then clicked edit and navigated to the Account Lockout Policy. From edit → Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Account Lockout Policies. I then clicked on the policies I wanted to enforce which I decided were all of them. I liked that there was a tab that explained what each policy did so I could learn more about it.



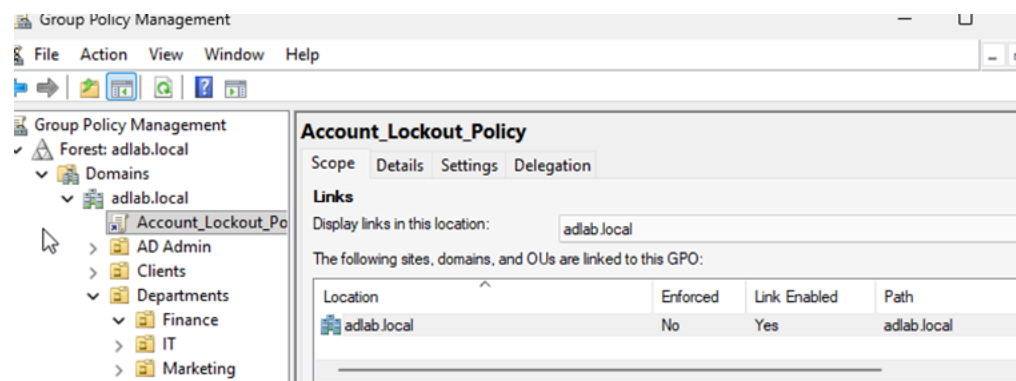
Next, I decided I wanted to define the Password Policy and set the maximum and minimum password age, password must meet complexity, and password length.



After I finished defining the password policies, I took the time to scroll and look through all of the different policies that you can define on Group Policy Management. I was shocked at how many policies you can define. It took a while to look through all of them because of the sheer amount. It could be the simplest thing like disabling a tab on the control panel and adding a "Logoff" button on the start menu, to installing binary behavior security restrictions and Mime sniffing safety features (Whatever that means). I researched some policies that were completely foreign to me and I was just blown away by how there was a vast range of policies that you could define. I realized it takes a whole lot of setup and management in order to securely manage end point users in a real corporate environment.



I then backed out and verified that my newly created GPO had all of the policies that I wanted. Next, I right clicked the domain name, then hit “Link an Existing GPO” and clicked on the GPO I had just created.



I wanted to test this by logging into a different user and trying to get my account locked out. To do that I created a whole new Client VM and used Jane Doe as my user. I then set it up and added it to the domain using the same process I did for the Michael Jackson user. On my DC, I moved Jane Doe’s computer into the IT OU which is the “department” that Jane Doe works in. Finally, I opened up the command prompt and ran the command “gpupdate /force” which I had learned to do during my internship. This command enforces the GPO to the user account.

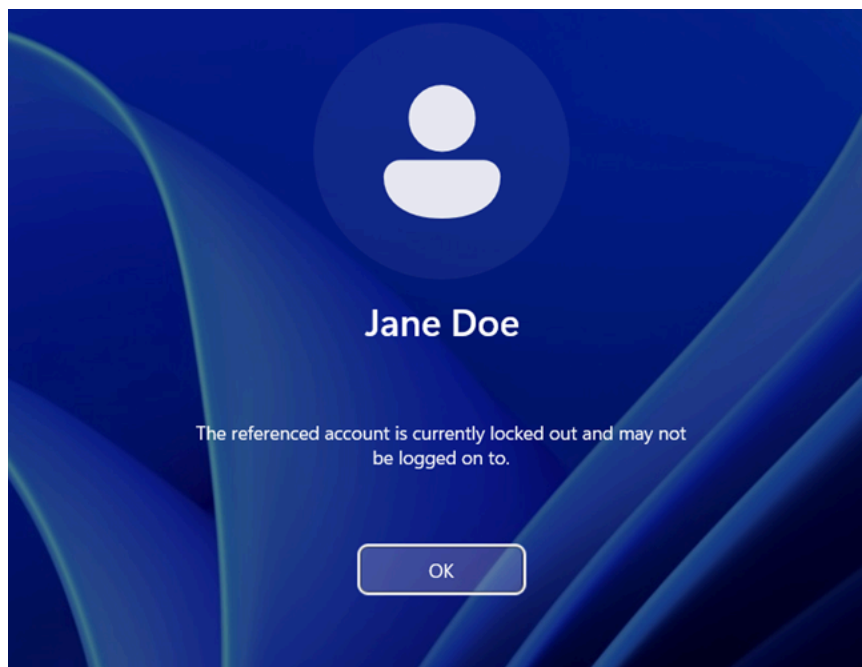

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\jdoe>gpupdate /force
Updating policy...

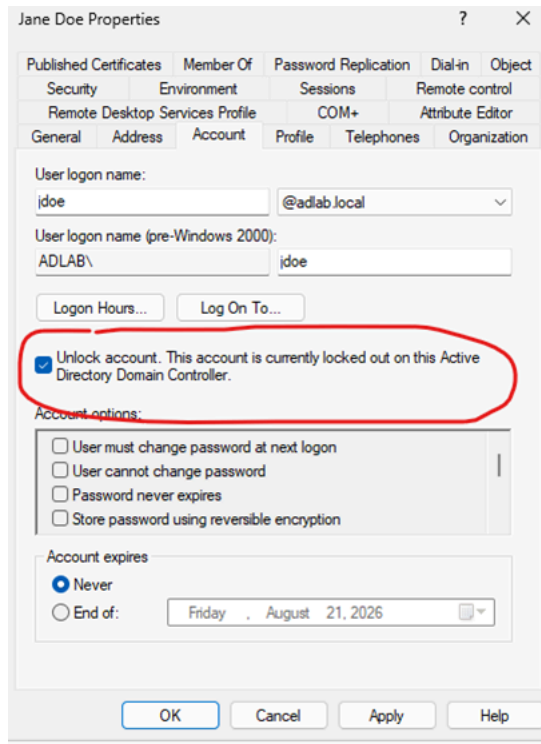
Computer Policy update has completed successfully.
User Policy update has completed successfully.

C:\Users\jdoe>
```

After 4 failed login attempts (as set by our GPO), we can see that Jane's account is locked!



Now that I confirmed that the GPO worked, I went back to AD and unlocked Jane's account.



I did do a little more GPO practice after writing this, but I wanted to at least show the first GPO I added to the domain. Overall, I thought Group Policy Management was a really powerful tool that allows you to better handle users in a workplace environment, following certain security and legal policies that may be established by a company.

Things I struggled with

I really struggled with setting up the domain controller. Even though I had Google to fall back on, I found myself getting stuck sometimes on what step to do next. I also struggled with Group Policy Management because it happened to not be as straightforward as I originally thought. I made a mistake when trying to set it up because I didn't properly link the GPO to the right computer. It took me a while to troubleshoot it because I wanted to try my best to figure out what the issue was before turning to my sources. Another step that really had me up late at night was the whole DHCP process. I have created a whole subsection for it because troubleshooting the issue took a lot of research and understanding. I would say the hardest part of this entire project was configuring the Domain Controller to have AD and DHCP on it, and troubleshooting all the mistakes I made along the way.

- Troubleshooting DHCP

After finishing the setup for the DHCP server the first time (creating and activating the IP scope), I didn't know that I had to do some extra steps to have the server send out IPs to every new client VM that joins the domain.

First, I did some research on what exactly was the issue and what I had to further configure in order for this to work. I first deleted the IP Scope so that I could restart the process. I then put in 192.168.10.50 - 192.168.10.200 as my scope and clicked activate. I learned that I had to configure the DHCP options by going to the DHCP server, right clicking Scope Options then clicking Configure Options. I then enabled "003 Router", "006 DNS Servers", "015 DNS Domain Name". I set the router to match the default gateway on the DC, then the DNS Server to match the DNS server of my DC (which was the same IP as the default gateway), and of course the DNS Domain Name as my Domain.

Next, I booted up my client VM and went to Control Panel → Network and Sharing Center → Change adapter settings → right clicking the Ethernet adapter → Properties → double clicking on IPv4 → selecting "Obtain an IP address automatically" and "Obtain DNS server address automatically", then "Configure" and "Yes".

Then I went to the command prompt and ran these 3 commands in order:

```
Ipconfig /release  
Ipconfig /renew  
Ipconfig /all
```

When I was first trying to diagnose the issue (before I had correctly set my DC to have a static IP), running "ipconfig /renew" would leave me waiting for a response for a while before I decided to do more research. After my DC VM had a good static IP, the "ipconfig /renew" command went through and I was finally able to use the DHCP server.

```
C:\Users\mjackson>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : adlab.local
    Link-local IPv6 Address . . . . . : fe80::7902:4a7a:634c:f713%9
    IPv4 Address. . . . . : 192.168.10.51
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.10.10
```

```
C:\Users\mjackson>ipconfig /release

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::7902:4a7a:634c:f713%9
    Default Gateway . . . . . :
```

```
C:\Users\mjackson>ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : adlab.local
    Link-local IPv6 Address . . . . . : fe80::7902:4a7a:634c:f713%9
    IPv4 Address. . . . . : 192.168.10.51
```

```
Link-local IPv6 Address . . . . . : fe80::7902:4a7a:634c:f713%9
IPv4 Address. . . . . : 192.168.10.51
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.10.10
```

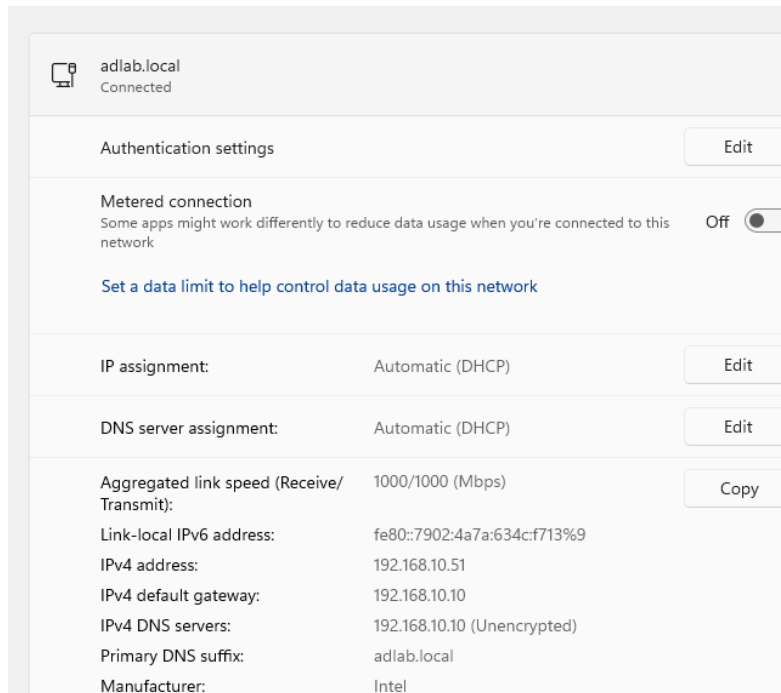
```
C:\Users\mjackson>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : MJACKSON-LAPTOP
    Primary Dns Suffix . . . . . : adlab.local
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : adlab.local
```

```
Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : adlab.local
    Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
    Physical Address. . . . . : 08-00-27-F0-4E-F0
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::7902:4a7a:634c:f713%9(Preferred)
    IPv4 Address. . . . . : 192.168.10.51(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Sunday, August 2, 2026 4:28:08 PM
    Lease Expires . . . . . : Monday, August 10, 2026 4:28:07 PM
    Default Gateway . . . . . : 192.168.10.10
    DHCP Server . . . . . : 192.168.10.10
    DHCPv6 IAID . . . . . : 84410407
    DHCPv6 Client DUID. . . . . : 00-01-01-00-31-D6-71-B1-08-00-27-F0-4E-F0
    DNS Servers . . . . . : 192.168.10.10
    NetBIOS over Tcpip. . . . . : Enabled
```



Things I've Learned

In addition to all the learning I did during my internship (working AD), I learned how to set up a domain controller VM and connect different Windows client VMs to the newly created domain. I also am starting to understand IP configurations, static IPs versus dynamic IPs, and setting up RRAS (Routing and Remote Access), and DHCP servers a little bit more. I believe that I gained a ton of knowledge on how to set up a Domain Controller with AD, DHCP, and DNS servers. Practicing some of the different features like creating Organizational Units and utilizing Delegation of Control allowed me to understand how to manage privileges of users in AD. After a long process of learning Group Policy Management, I now feel comfortable creating and linking GPOs to client's machines. In addition, I understand how to create secure end point users by creating different GPOs depending on my needs and linking them to specific OUs in an Active Directory domain.

Conclusion

Despite it being a bumpy road with a lot of roadblocks, this project really helped me understand how companies are able to securely manage end point users. It also helped me develop a mindset where I must first find the cause of the issue, understand what I'm dealing with, then find ways to troubleshoot and solve that issue without breaking anything else. Overall, I was excited to start, work through, troubleshoot, and complete this project and now that I can say that I'm comfortable doing some of these things, it makes me really excited for what the IT world has to offer.

References

<https://www.microsoft.com/en-us/software-download/windows11>

<https://www.microsoft.com/en-us/evalcenter/download-windows-server-2025>

<https://community.spiceworks.com/t/group-policy-best-practices/1012678>

<https://www.youtube.com/watch?v=jolubWzQ6P8>

ChatGPT, Gemini, and Reddit to explain certain things that I needed to configure on the DC VM (e.g., Routing and Remote Access, DHCP, installing AD).

<https://medium.com/@chryb3r/setting-up-an-active-directory-lab-using-virtual-box-87aef88e76ba>

(only looked at to keep me on track and figure out how to install AD using the Server Manager.)